11. MAP COLORING FOR CSP

Aim:

To implement the Map Coloring Problem using Constraint Satisfaction Problem (CSP) in Python.

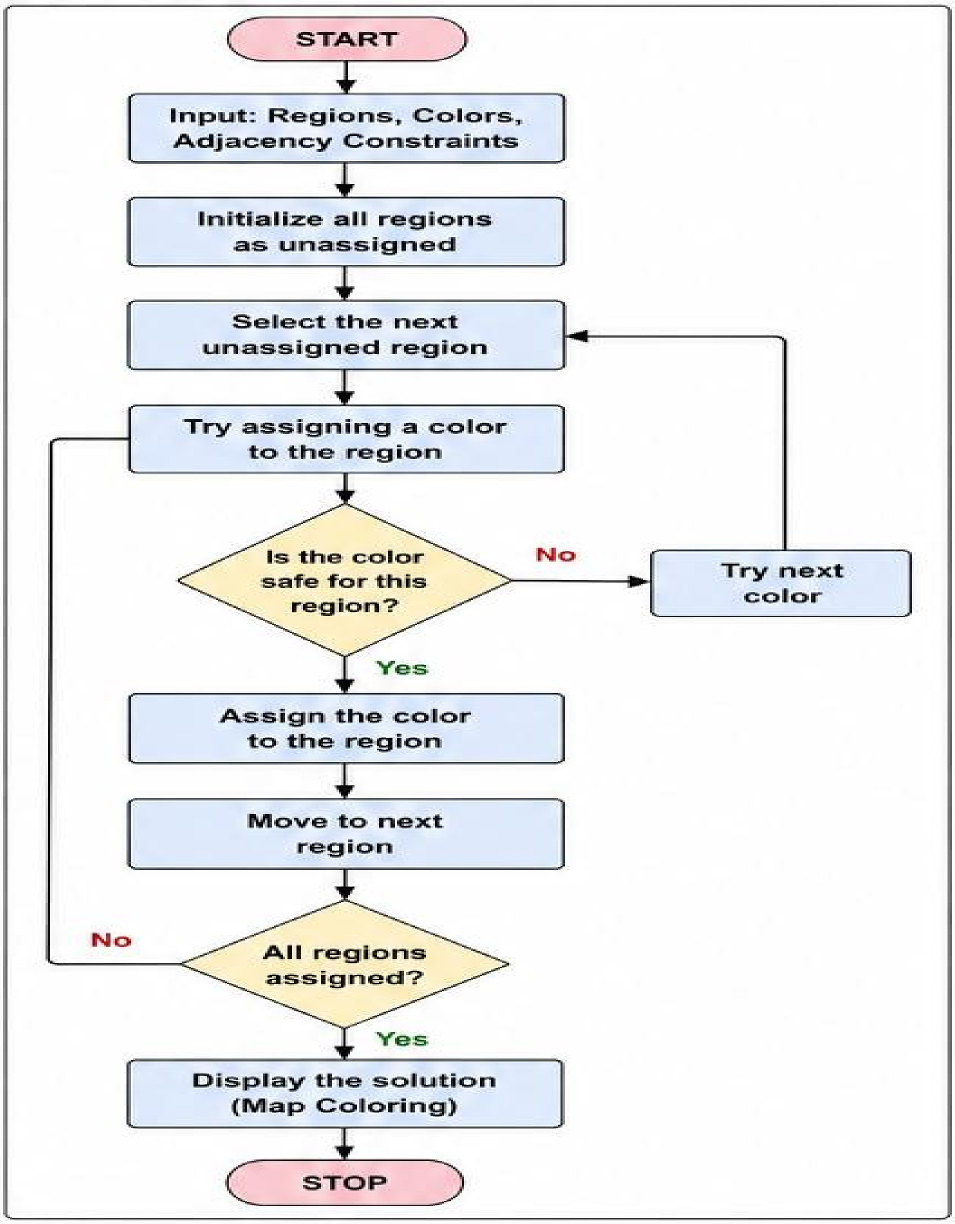
Objective :

To assign colors to different regions of a map such that no two adjacent regions have the same color.

Algorithm:

1. Define the regions and their neighboring relationships.
2. Define the available colors.
3. Select an unassigned region.
4. Assign a possible color to the region.
5. Check whether the assignment satisfies all neighboring constraints.
6. If valid, continue with the next region.
7. If invalid, try another color.
8. If no color is possible, backtrack to the previous assignment.
9. Continue until all regions are colored.
10. Display the valid coloring.

Flowchart :



Code:

```
regions = ["A", "B", "C", "D"]

colors = ["Red", "Green", "Blue"]

neighbors = {

    "A": ["B", "C"],

    "B": ["A", "C", "D"],
```

```
    "C": ["A", "B", "D"],
    "D": ["B", "C"]
}
def is_valid(region, color, assignment):
    for neighbor in neighbors[region]:
        if neighbor in assignment and assignment[neighbor] == color:
            return False
    return True
def map_coloring(assignment):
    if len(assignment) == len(regions):
        return assignment.copy()
    for region in regions:
        if region not in assignment:
            break
    for color in colors:
        if is_valid(region, color, assignment):
            assignment[region] = color
            result = map_coloring(assignment)
            if result is not None:
                return result
            del assignment[region]
    return None
solution = map_coloring({})
print("MAP COLORING USING CSP")
print("----------------------")
if solution:
```

```
    print("\nValid Coloring:")
    for region in regions:
        print(region, "->", solution[region])
else:
    print("No valid coloring exists.")
```

Output:

```
Output

MAP COLORING USING CSP
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Valid Coloring:
A -> Red
B -> Green
C -> Blue
D -> Red

=== Code Execution Successful ===
```

Result :

The Map Coloring Problem was successfully implemented using Constraint Satisfaction Problem (CSP) with backtracking. A valid color assignment was obtained such that no two adjacent regions have the same color.